

Luv Gupta

+91-9958493575 | luvgupta1805@gmail.com | linkedin.com/in/luv-gupta1800 | github.com/Luv1881

EDUCATION

Vellore Institute of Technology, B.Tech. in Computer Science (InfoSec) Sept. 2022 – May 2026

- **CGPA: 9.04/10**
- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Computer Architecture, Penetration Testing, Information Security, Web Application Security, Watermarking and Steganography
- Earned the **AWS Certified Cloud Practitioner Certificate** [link]

TECHNICAL SKILLS

- **Languages:** C++, Python, Java, JavaScript, SQL, Bash
- **Security & DevOps:** STRIDE, OWASP Top 10, CodeQL, Threagile, GitHub Advanced Security, Docker, GitHub Actions
- **Frameworks & Tools:** Boost.Asio, Boost.Beast, Flask, NextJS, Node.js, React, Firebase, AWS CLI, Git
- **Libraries:** OpenSSL, RE2, SQLiteCpp, spdlog, TensorFlow, Scikit-learn, NLTK

EXPERIENCE

SolarWinds India, Software Engineering Intern Jan. 2026 – June 2026

- Built a **C++20 reverse-proxy WAF** using Boost.Asio — async I/O, zero-downtime policy reload via atomic shared_ptr swaps, mTLS policy sync.
- Automated GitHub secret and code scanning triage with severity tiering and false-positive filtering, significantly reducing manual review effort.
- Pitched automated threat modeling across **STRIDE, VAST, PASTA** and other methodologies by extending Threagile, proposing it as a standardized security review workflow.

Datawise, Intern at Founder's Office May 2025 – July 2025

- Designed a fault-tolerant on-premise network architecture with HAProxy, Proxmox clusters, ZFS replication, and dual-ISP failover, reducing projected downtime risk by **40%**.
- Evaluated 10+ enterprise routers and firewall appliances; delivered a production-ready architecture diagram adopted for deployment.

Tech Mahindra, AWS System Operations Intern May 2024 – July 2024

- Rightsized EC2/S3/Lambda resources, cutting monthly AWS spend by **15%**; automated 10+ deployment tasks via AWS CLI, reducing release cycle time by **30%**.
- Built CloudWatch dashboards and alerting across multiple services, enabling proactive monitoring and faster incident response.

PROJECTS

Asio-Firewall | C++20, Boost.Asio, Boost.Beast, OpenSSL, SQLite, ZeroMQ

- Engineered a high-performance reverse-proxy WAF with async HTTP parsing, regex-based rule evaluation (RE2), IP filtering, and rate limiting — handling **10K+ concurrent connections** on a single node.
- Implemented a dual-plane architecture: data plane for live traffic inspection and control plane for ZeroMQ-based policy distribution with SQLite persistence and lock-free cache refresh.

Steganography & Watermarking Tool | Python, OpenCV, NumPy, PIL

- Implemented LSB-based image steganography to embed and extract hidden payloads in PNG/BMP files with imperceptible visual distortion (PSNR >**50 dB**); extended to DCT-domain watermarking for tamper detection.

PillBox | NextJS, Firebase, Flask, Git | link

- Led development of a medication delivery system for the elderly, improving accessibility by **30%**; integrated Firebase for real-time data handling, user auth, and cloud storage.

POSITIONS OF RESPONSIBILITY

Project Head, IET-VIT Sept. 2023 – May 2024

- Led technical events including Sapient Synthesis (300+ attendees) and ML 301; delivered hands-on sessions in ML model development and **penetration testing fundamentals** for 120+ students.